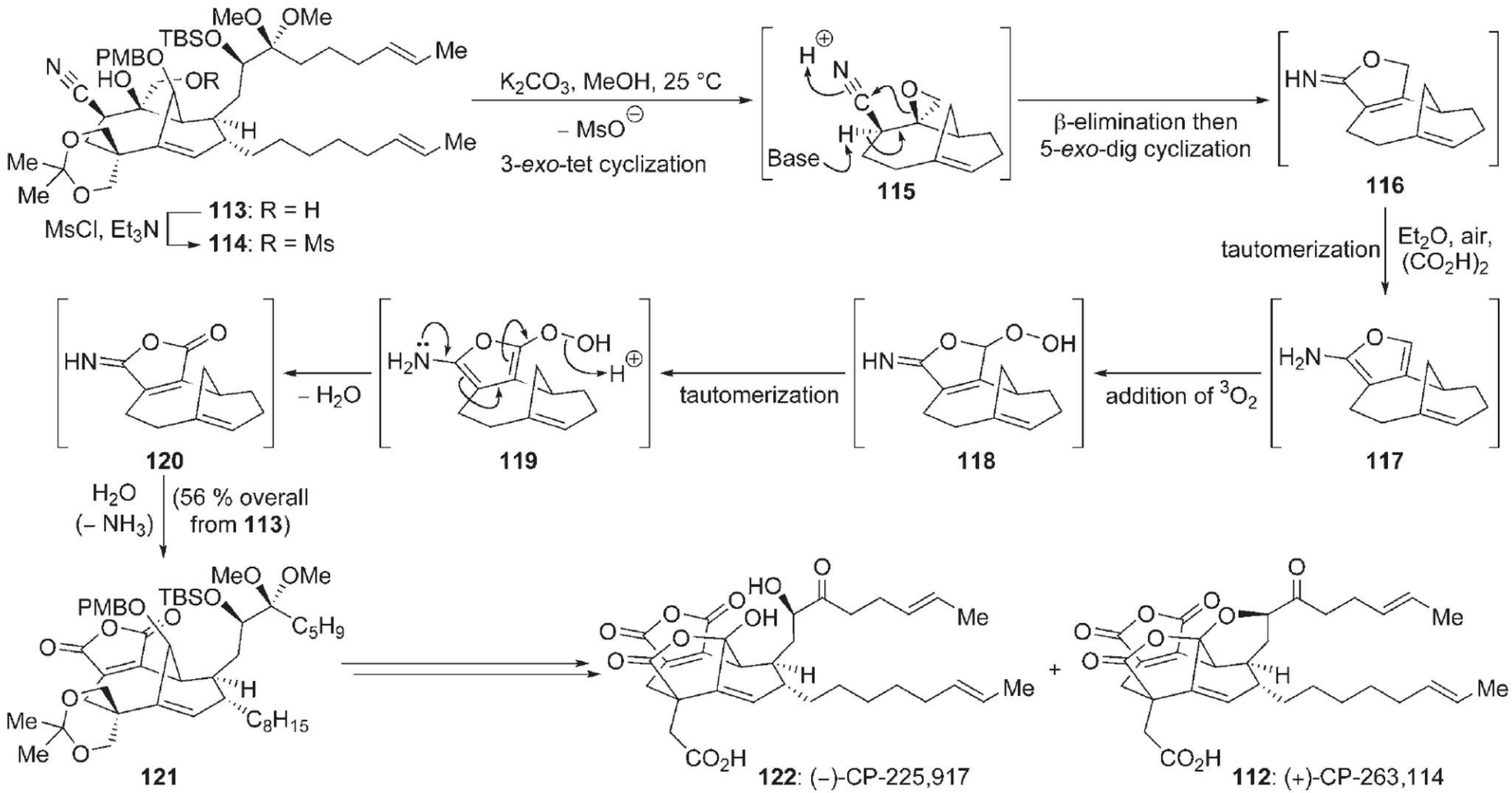

PhMe2Si
R
CN
O
1
OH
OMe
Me
13a
(R = CO2Et)
a) LiBH4 (1:1 d.r.)
b) p-TsOH, 14
c) DMP, NaHCO3
70% (3 steps)
MeO
OMe
14
X
CN
O
O
O
OMe
Me
15
d) NaHMDS
Ph3PCH3Br;
then HCl
84%
(X = SiMe2Ph)
OH
OH
OMe
16
e) PhI(OAc)2, NaHCO3
f) mesitylene, 170 °C
(X = SiMe2Ph)
17a (1-βOH, 33%)
17b (1-αOH, 36%)
1) DMP
2) NaBH(OMe)3
69% (2 steps)


MeO
OMe
TBSO
PMBO
HO
OR
Me
H
Me
Me
O
MsCl, Et3N
113: R = H
114: R = Ms
K2CO3, MeOH, 25 °C
– MsO⊖
3-exo-tet cyclization
Base
115
β-elimination then
5-exo-dig cyclization
116
tautomerization
Et2O, air,
(CO2H)2
117
addition of 3O2
118
tautomerization
119
– H2O
120
H2O
(– NH3)
(56 % overall
from 113)
C5H9
C8H15
121
HO
CO2H
122: (–)-CP-225,917
+
112: (+)-CP-263,114